

Software Defect Prediction Using Machine Learning Techniques

C.Lakshmi Prabha
Computer Science and Engineering
Thiagarajar College of Engineering
Madurai, India
lakshmiprabha@student.tce.edu

Dr.N.shivakumar
Computer Science and Engineering
Thiagarajar College of Engineering
Madurai, India
shivaa@tce.edu

Abstract—Software defect prediction provides development groups with observable outcomes while contributing to industrial results and development faults predicting defective code areas can help developers identify bugs and organize their test activities. The percentage of classification providing the proper prediction is essential for early identification. Moreover, software-defected data sets are supported and at least partially recognized due to their enormous dimension. This Problem is handled by hybridized approach that includes the PCA, randomforest, naïve bayes and the SVM Software Framework, which as five datasets as PC3, MW1, KC1, PC4, and CM1, are listed in software analysis using the weka simulation tool. A systematic research analysis is conducted in which parameters of confusion, precision, recall, recognition accuracy, etcAre measured as well as compared with the prevailing schemes. The analytical analysis indicates that the proposed approach will provide more useful solutions for device defects prediction.

Keywords—Defect prediction softwares; machine learning methods; metric softwares; prediction defect model; quality software;

I. INTRODUCTION

In the past decade, humans have progressively focused on software-based systems in which software quality is regarded as the most critical element in user functionality. Because of the vast production of application software, software quality remains an unresolved problem that gives inadequate output for industrial and private applications. Designs of defect prediction are commonly utilized by industries and Such models help in predicting faults, estimating effort testing software reliability, hazard analysis, etc. during the growth stage. A supervised machine learning predictive algorithm is consumed with the predefined collection of training data. The algorithm then gains expertise from the training dataset and produces rules for predicting the class label for a new data set. Learning phases consists to use mathematical algorithms to generate and strengthen the predictor function. Training data used in this process has an attribute input value and its defined output value. The expected ML algorithm quality is compared with the often known output. This is repeated in many iterations of training data until the optimal prediction accuracy is reached or the upper limit number of loops is finished. In the field of unsupervised learning algorithms, the class label output value is not known in data. Alternatively, a cluster of data loads the software, and the algorithm identifies a pattern and relationships within it. The main emphasis is on the

Relationship among data attributes. An example could be to define a group of friends on a website for social networking. Software quality can be enhanced by predicting defect modules. Defect prediction is the method of designing models that are utilized in the initial stages of the process to detect defective systems such as units or classes. This can be achieved by classifying the modules as defect prone or not. Different methods are used to identify the classification module, the most common of which is support vector classifier (SVC), random forest, naïve bayes, decision trees (DT), neural networks (NN). The detected defect prone modules are given high priority in progress testing phases and the non-defect prone modules are examined as time and cost permits. The feature of classification, known as the relationship between the attributes and the training dataset class label is established on the classifier method and examined through the formulae for the categorization of the targets. Those rules are also needed to define future dataset class labels. Thus, the unknown datasets can be categorized using the classification patterns and a classifier. Defining software defects, finding the defect and identifying it is a repetitive work for researchers due to the massive deployment of software. The main goal of categorizing the software dataset as a model for bug prediction into a defective and non-defective dataset. The input software dataset is given to the classifier according to this method where the user knows the actual class values. Requirement-based and design-based metric methods demonstrated considerable results before this scheme. But the design of algorithms and the accuracy of predictions remain a problem able task.

RELATED WORK

Machine learning is a powerful methodology for prediction, software defect prediction model proposed by Wang et al. [3] for increasing the quantity of application software systems. Databases of defective software comprise of unbalanced data which produces random patterns. This problem encourages the creation of an effective and reliable classifier of situations for academic and industrial applications. Xu et al. [4] researched “software defect prediction strategies and hypothesized that traditional techniques use vectorization and feature selection” framework to minimize trivial features, but still exclude other essential features resulting in degraded performance of defect prediction strategy. A piece of maximum information, data

correlation-based technique is proposed to tackle this problem. Recently, Duksan et al.[5] discussed the unbalanced nature of software defect results, and very few occurrences display attributes that belong to the defective class during the prediction process. This phase creates a reduction of efficiency in software industries and therefore involves a specific classification scheme. To resolve this problem, the whole issue is transformed into an issue of multi-objective optimization, where a Multi goal system of learning is implemented by analyzing a varied cross-project environment. Shan et al.[6] “utilized a well-known methodology in machine learning, i.e. SVM (support vector machine). Besides, predictability in attributes is discussed through the diligence of a locally linear embedding strategy with a support vector classifier. SVM constraints are indeed configured with a tenfold cross-validation process and grid search scheme according to this approach”. Experimental analysis reveals that the LLE-SVM works well for detecting defects. Yang et al.[7] “proposed the Predicting Software Deficiencies using a neural network method in which the neural network concept is incorporated along with the Bayesian approach as a radial basis. The efficiency of the radial neural network can be enhanced by optimizing the weight update framework, using a single Gaussian and two Gaussian structures, while the motivation-minimization scheme is often employed for weight realization”. Han et al., [8] “as stated in the proposed software development based stable program quality estimation model. Our approach involves an advanced software reliability template, a system building forecast model, a Rayleigh model, and a computer-assisted software safety estimate to boost predictive results”. Parthipan et al.[9] “have presented an analytical model describing the signs of design uncertainty using an aspect-oriented approach for measuring uncertainty. Noticed that defect prediction models are mainly developed in the design phase and code level either to differentiate between unreliable and non-faulty (binary classification) or to estimate the number of defects (regression analysis)”. Panichella et al.[10] “enhanced the recognition of defect-prone instances in software projects through a unified predictor of defects that brings into consideration the clusters provided by different approaches of machine learning”. Felix et al.,[2] have “proposed a study for the prediction of software defects using a machine learning method focused on the neural network. Github databases are regarded in this work for the study of defect prediction. A NN is applied with the aid of registry relationships between software codes and their faults, and to obtain classification and prediction. In classification and prediction strategy based on machine learning, feature selection and reduction will increase performance. By referring to that as a significant aspect”. Lu et al. [12] “used a version of the algorithm for self-study, to examine the implementation of a semi-supervised learning technique for software defect prediction. The research concluded that trust fitting could be used as a replacement for existing supervised algorithms. In conjunction with dimensional reduction, the semi-supervised algorithm behaved significantly better than a random forest model when training modules with typical defects were used”. Shepperd et al.[13] “carried out a meta-study of all the factors affecting output in predictions. As calculated based on the Matthews correlation coefficient, they checked the efficiency

of their defect prediction system by evaluating the conditions that strongly impact the predictive results of the software defect classificatory. They noticed that classifier choice affects output only marginally, while model building factors (i.e. Factors specific to the study group) produce a major impact. This is since the group of researchers is in charge of pre-processing the data”. Jayanthi et al.,[1] “established a Selection of features for apps scheme the unequal dataset of software defects. Next, the selection with attributes built on the wrapper is implemented, culminating in the collection of subsets of attributes. In the next process, random sampling is implemented to help reduce the negative impact of the unbalanced dataset”.

II. RESEARCH METHODOLOGY

The collection has been made to include the most common and often used machine learning methods. The following techniques are listed combined with their textual explanations.

A. Naïve Bayes

Naive Bayes is an applied training that is used for statistical technique knowledge grouping. When the name implies Naive this approach casually claims the attributes of a specific class are autonomous. Features assume powerful or naive isolation. It functions as a template applied to problematic objects as class labels, allocated as a vector used to draw class descriptors from finite sets. Naive bays are categorized into the complex situation of real-world problems given their simplistic nature and assumptions.

B. Random Forest

The purpose of the algorithm is to create a framework capable of predicting value function based on different input factors. Each internal node reflects one of the input parameters. For each potential value of these factors, there are boundaries for their offspring. Leaf in the tree describes the goal factor in which the specified parameters of the input factor can be crossed by way of the root to the leaf. The learning strategy uses the random forest as a statistical model where the target quality interpretation of the item is mapped with an analysis of the object. This is a predictive technique used in mining, statistics, machine learning.

C. SVC

The learning method allows for interpreting the information utilized in the categorization and regression analysis. An SVM model is defined as test samples in the range that are distributed and in that way that they are divided by a gap far as feasible based on the divisions to which they belong. The classification of new samples is calculated per the side of the gap to which they fell after mapping into a certain area.

An SVM constructs a collection of hyperlens in a non-dimensional space used in correlation, classification. A hyperplane with the maximum distance to the set of points in a particular class termed functional margin. Ultimately, the functional margin was in inverse proportion to the error in generalization.

D. Neural Network

The assigned as the network of connections between the neurons that can share information to connect. The working of an ANN is defined as follows. First, the neural network accepts the values of the data variables as an input node of the input layer. Weights are allocated to the ties that bind nodes. These numerical Weights are balanced according to the NN able to learning and adapt. Nodes are crossed and the values of the variables are determined to move through the network. The weight of each connection affects the result of the parameter value. At the output node, the parameter value is matched with the target value and the impact is expected.

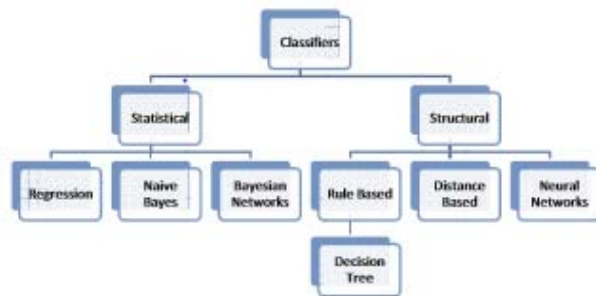


Fig 1. Machine learning techniques Classification

III. PROPOSED APPROACH

Prediction of software defects is a critical task in the area of software engineering. The prior chapter explains methods of prediction of defects centered in the machine learning software. Moreover, these methods pointed at software bug mismatch challenges, but also classification accuracy and overall efficiency stay a difficult task for researchers. To fix this problem, a hybrid feature reduction scheme is presented and artificially dependent neural network strategy for the prediction of software defects. The first subparagraph of this article comprises an enhanced PCA method for dimensional reduction and numerical modeling, while the second subparagraph collects data on the combined application of the neural network and the current PCA method.

A. PCA

PCA stands for principal component analysis. PCA is a method for limiting the size of these datasets, increasing their computational complexity while at the same rate minimizing information loss. Primary component analysis is a computational method, PCA's goal is to reduce the dataset's dimensionality. It is sometimes termed a linear orthogonal transformation turning the data into a new quaternion. The key problem is that PCA is a strategy for extracting items, rather than a tool for selecting features. The linear variation of initial characteristics is producing new attributes. For executing the reduction the features with the lowest variance are appropriate. Many papers like those used PCA to improve the efficiency of their experimentations. "The PCA method, according to this, converts n vector $\{x_1, x_2, \dots, x_n\}$ from d -dimensional space into n vectors $\{x_1, x_2, \dots, x_n\}$ in a new dimension space".

The advantage of the proposed approach is that the feature collection methodology recognizes and selects the data set's

most important functionality for learning purposes, the features are utilized more in predicting. The complexity in the information is reduced and the efficiency of the learning method can be enhanced. Input data play an important task in categorizing and predicting results.

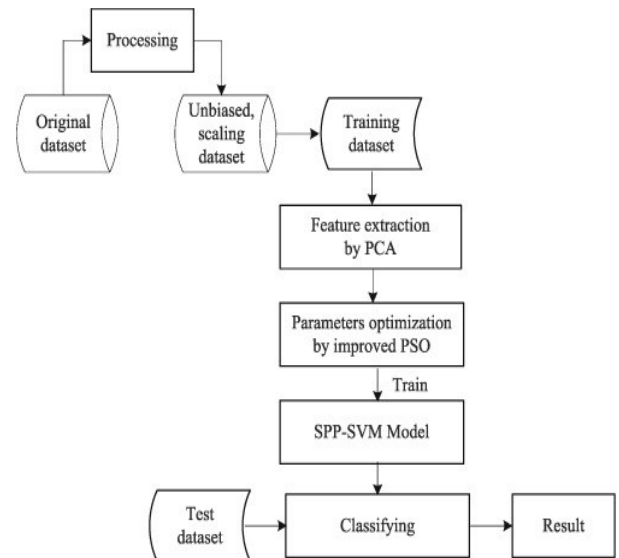


Fig 2. Software Defect Prediction Model

PSO stands for Particle Swarm optimization. An optimal solution to the problem is identified in this method by enhancing the quality of the candidate solution which is an initial solution. every candidate in this algorithmically is known locally as particle. this particle navigates the search area according to the to a system based on the particle position and velocity. Another parameter that depends on serialization is the best search space positions. PSO can find a solution (near to optimal) in broad find spaces. There is no guarantee that this algorithm will be used to seek the solution.

This algorithm has different implementations to illustrate diverse fields such as ANN, fuzzy controllers, and problems of optimization. Assume a circle of friends is looking and get out of the jungle and there was only one path out. they do not know the path to exit but understand only one parameter which is relative distance. Every person in this issue can be viewed as a particle and so these particles will be given a particle place and velocity. According to PSO, the speed of every individual particle depends on individual location to the exit point.

Certain parameters in PSO can be modified which allows everything more attractive. Mild variance in one implementation makes good for a huge variety of applications. Particle Swarm Optimization is being used in various fields where technologies are very large and even unique applications that concentrate on specific criteria.

B. DataSet

There are several free access databases available on the internet. Kaggle promise dataset database. Five datasets namely KC1, PC3, PC4, MW1, CM1 supported the data sets.

Data retrieval (referred to as AT in this survey) and numeric defect level KC1 classes were utilized.

Throughout this study, addressed Kaggle PROMISE database datasets which are called KC1, CM1, PC3, MW1 and PC4 where different attributes are present in the given dataset. The tables show various metrics about the assumed dataset, showing a attributes count , usable components, faulty components, and defective percentage.

Through these datasets, a software deficiency prediction

Dataset	Precision	Recall	F1-measure	Support	Acc
KC1	T-1.00 F-1.00	F-1.00 T-0.99	F-1.00 T-0.99	F-438 T-90	1.00
CM1	T-0.99 F-1.00	N-1.00 Y-0.88	N-1.00 Y-0.93	N-117 Y-8	0.99
PC4	N-1.00 Y-1.00	N-1.00 Y-0.94	N-1.00 Y-0.97	N-260 Y-18	0.96
MW1	N-0.89 Y-0.87	N-0.89 Y-0.89	N-0.96 Y-0.93	N-55 Y-9	1.00
PC3	N-0.87 Y-1.00	N-0.97 Y-0.94	N-1.00 Y-0.92	N-32 Y-238	0.97

method is used in which the output of the current proposal is compared to other art of state methods. Numerous measuring metrics “such as true positive, rate, false-positive rate, precision, uncertainty matrix, precision, and recall, etc. are present”.. This matrix includes the real and expected class value and can be performed based on those classification tests.

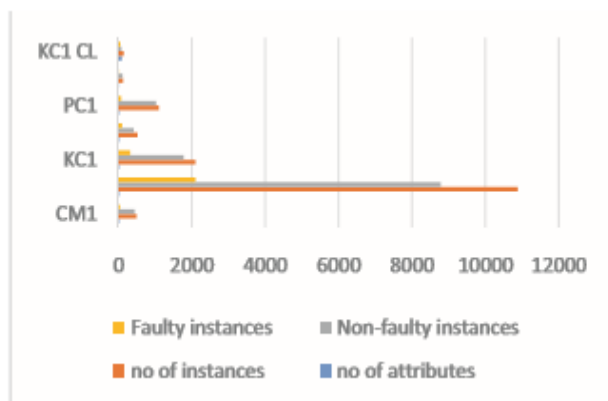


Fig 3.Characteristics of Dataset

The above chart shows the data sets of reference to the number of the flawed, non-faulty, actual number of qualities of each type of case.

IV. RESULTS AND ANALYSIS

The data sets were collected from the Kaggle promise dataset database. Five datasets are utilized, namely KC1, PC3,

PC4, MW1, and CM1. The algorithms chosen for analysis are Naive Bayes, Random Forest, SVC (Linear Regression), Neural Network.

The data sets are collected in arff form from the Kaggle database, assisted by the r studio tool. Therefore the data processing section mast was run to render data sets consistent with it. The outcome test summary is shown in the table below. It shows the accuracy value of each method (percentage accordance). The maximum heuristic in a dataset is labeled prominent to imply that amongst others.

TABLE I. PERFORMANCE EVALUATION FOR SVC

As it is evident from the tests that the linear classification technique has the greatest accuracy of prediction of defects in four of the five identified datasets, this is the most accurate methodology due to its higher precision. The remaining three algorithms were naive Bayes, random forests and neural networks with the maximum precision in one dataset.

TABLE II. PERFORMANCE EVALUATION FOR RANDOM FOREST

Dataset	Precision	Recall	F1-measure	Support	Acc
KC1	T-0.96 F-0.75	F-0.99 T-0.88	F-1.00 T-0.97	F-438 T-90	0.99
CM1	T-0.88 F-0.92	N-0.85 Y-0.93	N-0.92 Y-0.76	N-117 Y-8	0.98
PC4	N-0.95 Y-0.86	N-1.00 Y-0.93	N-0.97 Y-0.79	N-260 Y-18	0.98
MW1	N-0.96 Y-0.85	N-0.94 Y-1.00	N-0.85 Y-0.84	N-55 Y-9	0.97
PC3	N-0.94 Y-0.84	N-0.85 Y-0.95	N-0.89 Y-0.99	N-32 Y-238	1.00

The Above table displays the standard deviation loss description of each method. It exposes incorrect positive performance of each method (percentage terms). The maximum algorithm in the dataset is labeled highlight to show amongst several methods.

TABLE III. PERFORMANCE EVALUATION FOR NAÏVE BAYES

The table above describes the standard deviation of that same test from the predicted expected defect. The lowest failure rate is the neural network method. The rate of failure would help to overcome the tie. The lesser error, the greater the accuracy in the scenario of a tie between two algorithms between terms of prediction of defects.

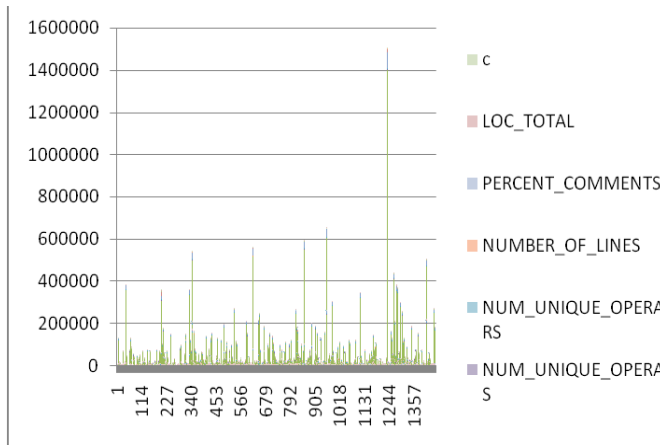


Fig 4. Prediction of defect accuracy in each dataset

Tdata denotes the training datasets that are used to construct our predictive systems, and Vdata denotes the testing datasets. The results were placed in Test, and the prediction was displayed in Pred.

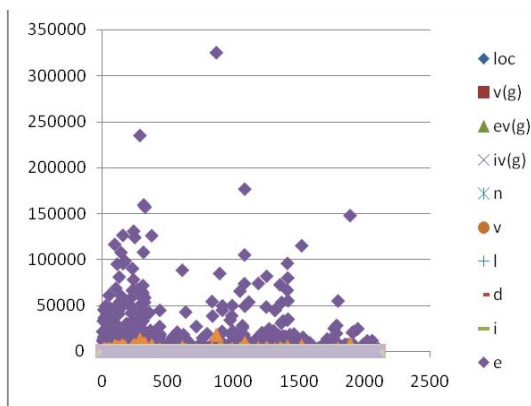


Fig 5. Standard Deviation for each Dataset

From the analysis, it is evident that perhaps the neural network has the least failure rate in the study preceded by random forest. Furthermore, the largest accuracy value is a linear classification (SVC). In the case of a dispute in accuracy estimation, error rate parameters could be considered to evaluate which method best performs.

V. CONCLUSION AND FUTURE WORK

This research primarily seeks to use information-mining techniques to predict software-defects. Moreover, this domain is now a significant research field whereby numerous strategies have been explored to somehow enhance the efficiency of detecting software defects or predicting bugs. Throughout this study, by designing a new hybrid model using

Dataset	Precision	Recall	F1-measure	Support	Acc
KC1	T-0.99 F-1.00	F-1.00 T-0.88	F-1.00 T-0.93	F-117 T-8	0.96
CM1	T-0.98 F-0.86	N-0.96 Y-0.93	N-0.97 Y-0.89	N-104 Y-27	0.97
PC4	N-0.98 Y-0.86	N-0.96 Y-0.93	N-0.97 Y-0.89	N-109 Y-16	1.00
MW1	N-0.92 Y-0.96	N-0.94 Y-0.93	N-0.96 Y-0.98	N-131 Y-29	0.99
PC3	N-0.91 Y-0.86	N-0.93 Y-0.89	N-0.97 Y-0.81	N-32 Y-238	0.98

feature reductions and classification, dealt with the issue of classification accuracy for massive datasets. PCA is introduced to acquire a feature reduction template, where overall probability is also applied to minimize the data recovered by PCA. Besides, the neural network classification method is used for the detection of program bugs. Research study indicates that the proposed method offers good efficiency and achieves AUC as 98.70 percent that is a major change relative to certain the art of state models. To show objectively the relevance of the feature selection effect. There is no clear variation in classifier accuracy if there is no methodology for selecting features and whenever the strategies for selecting features are employed. There is a gap in classifier accuracy when there is no set of features and when the methods for selecting the function are being used. Therefore, it is observed that the period and space difficulty for the prediction of defects is decreased without affecting the prediction accuracy by using feature selection techniques.

These findings can be more improved with the use of several datasets. An increase in the number of datasets can enhance the findings. It is also possible to compare further techniques. The most common and widely used methods were incorporated into consideration in this study. It is expected that new methods would be shown in the future and will be used in the detailed analysis

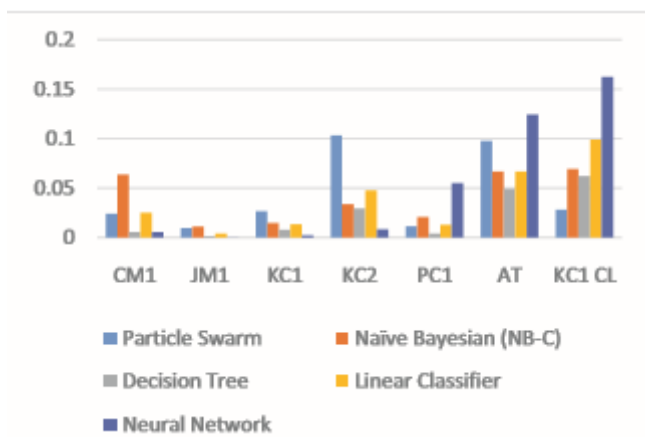


Fig 6.Overall Algorithm classification on Dataset.

The results indicate that perhaps the neural network has the least failure rate in the study preceded by random forest. The greatest detection rate, however, is the dimensional classification. In case of a prediction of tie accuracy, the failure rate parameter can be used to determine the correct outcome.

REFERENCES

- [1] Jayanthi, R. and Florence, L., 2019. Software defect prediction techniques using metrics based on neural network classifiers. *Cluster Computing*, 22(1), pp.77-88.
- [2] Felix, E.A. and Lee, S.P., 2017. Integrated approach to software defect prediction. *IEEE Access*, 5, pp.21524-21547.
- [3] Wang, T., Zhang, Z., Jing, X., Zhang, L.: Multiple kernel ensemble learning for software defect prediction. *Autom. Softw. Eng.* 23, 569–590 (2015).
- [4] Xu, Z., Xuan, J., Liu, J., Cui, X.: MICHAC: defect prediction via feature selection based on maximal information coefficient with hierarchical agglomerative clustering. In: 2016 IEEE 23rd International Conference on Software Analysis, Evolution, and Reengineering (SANER), Suita, pp. 370–381 (2016).
- [5] Ryu, D., Baik, J.: Effective multi-objective naïve Bayes learning for cross-project defect prediction. *Appl. Soft Comput.* 49, 1062 (2016).
- [6] Shan C., Chen B., Hu C., Xue J., Li N.: Software defect prediction model based on LLE and SVM. In: Proceedings of the Communications Security Conference (CSC '14), pp. 1–5 (2014).
- [7] Yang, Z.R.: A novel radial basis function neural network for discriminant analysis. *IEEE Trans. Neural Netw.* 17(3), 604–612(2006).
- [8] K. Han, J.-H. Cao, S.-H. Chen, and W.-W. Liu, “A software reliability prediction method based on software development process,” in *Quality, Reliability, Risk, Maintenance, and Safety Engineering (QR2MSE)*, 2013 International Conference on. IEEE, 2013, pp. 280–283.
- [9] S. Parthipan, S. Senthil Velan, and C. Babu, “Design level metrics to measure the complexity across versions of ao software,” in *Advanced Communication Control and Computing Technologies (ICACCCT)*, 2014 International Conference on. IEEE, 2014, pp. 1708–1714.
- [10] A. Panichella, R. Oliveto, and A. De Lucia, “Cross-project defect prediction models: L’union fait la force,” in *Software Maintenance*,
- [11] Bautista, A.M., Feliu, T.S.: Defect prediction in software repositories with artificial neural networks. In: Mejia, J., Munoz,M., Rocha,Á., Calvo-Manzano, J. (eds.) *Trends and Applications in Software Engineering.Advances in Intelligent Systems and Computing*, vol.405. Springer, Cham (2016).
- [12] H. Lu, B. Cukic, and M. Culp, “Software defect prediction using semisupervised learning with dimension reduction,” in *Automated Software Engineering (ASE)*, 2012 Proceedings of the 27th IEEE/ACM International Conference on. IEEE, 2012, pp. 314–317.